

TASK BRIEF

FIWARE feasibility research for the Kigali Command & Control Center

- Assigned by: CDO
- Priority: **High**
- Due: 2 weeks from assignment date

Background

The CoK digitalization division is evaluating open-standard data platforms to power Kigali's Command & Control Center (CCC), a unified city operations hub that consolidates real-time data from transport, inspection, urban planning, utilities, public safety, environment, and citizen services.

FIWARE, an open-source initiative backed by the European Commission, has emerged as a leading candidate due to its vendor-neutral architecture and proven adoption in smart city deployments globally. This research task requires a focused technical and strategic assessment of FIWARE's suitability for City of Kigali's specific infrastructure context.

Objectives

- **Understand the FIWARE ecosystem:** Core components, open standards (NGSI v2 and NGSI-LD), and how they support real-time city data management.
- **Identify CCC-relevant components:** Determine which FIWARE building blocks apply to a Command & Control use case: data ingestion, real-time dashboards, alerting, and system integrations.
- **Review reference implementations:** Study comparable city deployments (Santander, Antwerp, Buenos Aires, Cape Town) and extract lessons applicable to Kigali.
- **Assess technical feasibility for Kigali:** Evaluate infrastructure requirements, hosting options (cloud vs. on-premise), local engineering capacity, and connectivity constraints.
- **Propose a high-level architecture:** Sketch how FIWARE could serve as the data backbone for Kigali's CCC, integrating at least 3 city domains (transport, inspection, urban planning, environment, public safety).

Specific areas to research

CORE STACK

Orion Context Broker, NGSI-LD, QuantumLeap, Cygnus, Keyrock identity manager

SMART DATA MODELS

Transport, environment, waste, water, and public safety schemas at smartdatamodels.org

VISUALIZATION

Grafana, Kibana, or FIWARE Knowage as CCC dashboard layers

IOT INTEGRATION

FIWARE IoT Agents for sensors, CCTV, SCADA systems, and third-party city APIs

DEPLOYMENT OPTIONS

Docker/Kubernetes setup, self-hosting, cloud providers, and local data centre options

SECURITY & SOVEREIGNTY

Data sovereignty posture, access control via Keyrock + Wilma PEP Proxy, audit trails

Suggested resources

- Official step-by-step FIWARE tutorials for developers: <https://fiware-tutorials.readthedocs.io/en/latest/>
- FIWARE components catalogue: <https://www.fiware.org/catalogue/>
- FIWARE Smart City data catalogue: <https://smartdatamodels.org/>
- The dedicated NGSI-LD reference and API documentation site: <https://ngsi-ld.org/>
- All FIWARE open-source repositories: <https://github.com/FIWARE>
- SCEWC case studies and FIWARE city deployment white papers
- African Smart Cities Network publications for regional context

Deliverables

TIMELINE	DELIVERABLE
Week 1	Research summary document covering FIWARE architecture, relevant components, and an annotated list of smart city reference implementations with key lessons for Kigali.
Week 2	Technical feasibility assessment for Kigali + a proposed high-level architecture diagram showing FIWARE as the CCC data backbone + a presentation-ready slide deck (max 3 slides) for review.

Final Notes

Pay particular attention to data sovereignty implications as the City of Kigali requires a self-hostable or locally deployable solutions. Flag any licensing, cost, or skill-gap risks early in the Week 1 document so they can be addressed before the architecture proposal.